Social media app using python

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import os

# ------------------ USER CLASS ------------------
class User:
    def __init__(self, username, password, name):
        self.username = username
        self.password = password
        self.name = name

    def save_user(self):
        with open("users.txt", "a") as f:
            f.write(f"{self.username},{self.password},{self.name}\n")

    @staticmethod
    def login(username, password):
        if not os.path.exists("users.txt"):
            return None

        with open("users.txt", "r") as f:
            for line in f:
                u, p, n = line.strip().split(",")
                if u == username and p == password:
                    return User(u, p, n)
        return None

    def update_profile(self, new_name):
        self.name = new_name
        self._update_user_file()

    def change_password(self, new_password):
        self.password = new_password
        self._update_user_file()

    def _update_user_file(self):
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        users = []
        with open("users.txt", "r") as f:
            for line in f:
                u, p, n = line.strip().split(",")
                if u == self.username:
                    users.append(f"{self.username},{self.password},{self.name}\n")
                else:
                    users.append(line)

        with open("users.txt", "w") as f:
            f.writelines(users)


# ------------------ POST CLASS ------------------
class PostManager:
    def __init__(self, username):
        self.filename = f"posts_{username}.txt"

    def create_post(self, content):
        with open(self.filename, "a") as f:
            f.write(content + "\n")

    def view_posts(self):
        if not os.path.exists(self.filename):
            print("No posts yet.")
            return

        with open(self.filename, "r") as f:
            posts = f.readlines()

        for i, post in enumerate(posts, 1):
            print(f"{i}. {post.strip()}")

    def delete_post(self, index):
        if not os.path.exists(self.filename):
            print("No posts to delete.")
            return
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        with open(self.filename, "r") as f:
            posts = f.readlines()

        if 0 < index <= len(posts):
            posts.pop(index - 1)
            with open(self.filename, "w") as f:
                f.writelines(posts)
            print("Post deleted.")
        else:
            print("Invalid post number.")


# ------------------ APP CLASS ------------------
class SocialMediaApp:
    def __init__(self):
        self.current_user = None

    def register(self):
        username = input("Enter username: ")
        password = input("Enter password: ")
        name = input("Enter profile name: ")

        user = User(username, password, name)
        user.save_user()
        print("Registration successful!")

    def login(self):
        username = input("Enter username: ")
        password = input("Enter password: ")

        user = User.login(username, password)
        if user:
            self.current_user = user
            print(f"Welcome {user.name}!")
            self.user_menu()
        else:
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        print("Invalid credentials!")

    def user_menu(self):
        post_manager = PostManager(self.current_user.username)

        while True:
            print("\n1. Create Post")
            print("2. View Posts")
            print("3. Delete Post")
            print("4. Change Password")
            print("5. Change Profile Name")
            print("6. Logout")

            choice = input("Enter choice: ")

            if choice == "1":
                content = input("Enter post: ")
                post_manager.create_post(content)

            elif choice == "2":
                post_manager.view_posts()

            elif choice == "3":
                post_manager.view_posts()
                idx = int(input("Enter post number to delete: "))
                post_manager.delete_post(idx)

            elif choice == "4":
                new_pass = input("Enter new password: ")
                self.current_user.change_password(new_pass)
                print("Password updated.")

            elif choice == "5":
                new_name = input("Enter new name: ")
                self.current_user.update_profile(new_name)
                print("Profile updated.")
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        elif choice == "6":
            self.current_user = None
            print("Logged out.")
            break

        else:
            print("Invalid choice!")

    def run(self):
        while True:
            print("\n--- SOCIAL MEDIA APP ---")
            print("1. Register")
            print("2. Login")
            print("3. Exit")

            choice = input("Enter choice: ")

            if choice == "1":
                self.register()
            elif choice == "2":
                self.login()
            elif choice == "3":
                print("Goodbye!")
                break
            else:
                print("Invalid choice!")


# ------------------ MAIN ------------------
if __name__ == "__main__":
    app = SocialMediaApp()
    app.run()
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